Huawei Access Node YEL-ACC-22 Minor QoS Congestion Event Resolution SOP

Node: YEL-ACC-22 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_2761

Description

The QoS congestion event in the Huawei Access Node YEL-ACC-22 at Yelahanka Sub-station is caused by microbursts, interface tail drops, or QoS queue starvation, which can be attributed to the Node's inability to handle the bursty traffic. This is due to the Node's limited buffer size and the QoS policies not being optimized for the current traffic patterns. The Huawei Access Node, being a Layer 2 device, relies on the IEEE 802.1Q standard for VLAN tagging and prioritization, which can lead to queue starvation if not properly configured. Furthermore, the Node's default QoS policies may not be sufficient to handle the traffic congestion, leading to packet drops and microbursts. The Alarm Code ALARM_2761 indicates a minor QoS congestion event, which requires immediate attention to prevent network instability and potential data loss.

Pre-requisite Checks

- Ensure that the Node is stable and not experiencing any other critical issues.

- Verify that the redundant link is operational and not contributing to the congestion event.

- Check the Node's power supply and ensure it is functioning within the specified limits.

- Review the Node's temperature and humidity levels to ensure they are within the recommended ranges.

Potential Root Causes

Cause	Probability
Insufficient QoS policies to handle bursty traffic	High
Default QoS policies not optimized for current traffic patterns	Medium
Limited buffer size on the Node	Low
Interface configuration issues leading to queue starvation	Medium

Diagnostic Commands & Expected Outputs

```
Command: display current-configuration | include qos-profile, Expected Output: qos-profile profile1, Description: This command displays the QoS policies configured on the Node. The expected output includes the QoS profile name, which helps identify the policies in place. The description explains the command's purpose and the expected output, providing context for the troubleshooting process.

Command: display interface GigabitEthernet 0/0/0 | include queue, Expected Output: queue 10, Description: This command displays the
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interface configuration, including the queue settings. The expected
output includes the queue size, which helps determine if the interface
is configured to handle bursty traffic.

Command: display qos-traffic | include drop, Expected Output: dropped
packets: 100, Description: This command displays the QoS traffic
statistics, including the number of dropped packets. The expected
output helps identify if there is a QoS congestion event occurring on
the Node.
```

Resolution Steps

Step 1: Step: Optimize QoS policies for current traffic patterns, Command: config qos-profile profile1 priority 10, Description: This step configures the QoS policy to prioritize traffic based on the current traffic patterns. The command optimizes the QoS policies to handle bursty traffic, reducing the likelihood of queue starvation.

Step 2: Step: Increase buffer size on the Node, Command: config interface GigabitEthernet 0/0/0 buffer-size 100, Description: This step increases the buffer size on the Node, providing more space for bursty traffic. The command helps mitigate the effects of queue starvation by increasing the buffer size.

Step 3: Step: Update interface configuration to handle bursty traffic, Command: config interface GigabitEthernet 0/0/0 queue 10, Description: This step configures the interface to handle bursty traffic by increasing the queue size. The command ensures that the interface can handle the increased traffic, reducing the likelihood of queue starvation.

Post-Check Verification

```
Command: display qos-traffic | include drop, Expected Output: dropped
packets: 0, Description: This command verifies that the QoS congestion
event has been resolved. The expected output indicates that no packets
are being dropped, confirming that the QoS policies are optimized for
the current traffic patterns.

Command: display interface GigabitEthernet 0/0/0 | include queue,
Expected Output: queue 10, Description: This command verifies that the
interface is configured to handle bursty traffic. The expected output
confirms that the queue size has been increased, ensuring that the
interface can handle the increased traffic.
```

Roll-back Procedures

- Step: Restore original QoS policies, Command: undo config qos-profile profile1 priority 10, Description: This step restores the original QoS policies, reverting to the previous configuration. The command ensures that the Node returns to its original state, mitigating any potential issues caused by the resolution steps.

- Step: Decrease buffer size on the Node, Command: undo config interface GigabitEthernet 0/0/0 buffer-size 100, Description: This step decreases the buffer size on the Node, reverting to the previous configuration. The command ensures that the Node returns to its original state, mitigating any potential issues caused by the resolution steps.

- Step: Update interface configuration to handle normal traffic, Command: undo config interface GigabitEthernet 0/0/0 queue 10, Description: This step updates the interface configuration to handle normal traffic, reverting to the previous configuration. The command ensures that the interface returns to its original state, mitigating any potential issues caused by the resolution steps.